



COLLEGE OF INFORMATION TECHNOLOGY AND ENGINEERING
SY 2025-2026 | Second Semester
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

IskolarBlock: A Blockchain-Enabled Web Platform for Transparent Barangay Scholarship Processing

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ABSTRACT

This paper presents the design, implementation, and evaluation of IskolarBlock, a web platform system that leverages Optical Character Recognition (OCR) and blockchain hashing to improve transparency, integrity, and efficiency in barangay-level scholarship processing. Developed using an Agile Scrum approach, the system integrates secure OTP authentication, automated document extraction (ID, COR, COG), role-based admin controls, and blockchain logging of key transactions. Acceptability and software quality were measured through structured manual testing across three iterative cycles and a user evaluation based on the ISO/IEC 25010:2023 quality model. Results from 51 respondents and 76 test cases showed progressive improvement across cycles (82.89% → 94.74% → 98.68% pass rates) and “Strongly Agree” ratings across all ISO quality attributes, indicating the system is reliable, usable, secure, and ready for deployment.

KEYWORDS:

blockchain, OCR, scholarship management, barangay, software quality, ISO/IEC 25010:2023

INTRODUCTION

1.1 Background of the Study

Scholarships are a cornerstone of educational equity in the Philippines, yet the systems delivering this aid at the barangay level are often hindered by systemic inefficiencies. Current processes, from application to eligibility verification, remain largely manual and fragmented, resulting in processing delays, redundant tasks, and operational bottlenecks. Research indicates that traditional scholarship administration methods lead to processing delays, duplicate applications, and increased vulnerability to fraud, ultimately reducing timely assistance and transparency in local governance (Azzam et al., 2023; Abubo et al., 2024).

To address these structural flaws, blockchain technology offers a compelling framework. Characterized by its decentralized ledger and immutability, blockchain ensures that transactions are recorded in a verifiable chain, guaranteeing data integrity and transparency among stakeholders (Mohammed M.I, 2024; Azzam et al., 2023).

Unlike traditional databases, this technology provides an architecture where trust is programmatic rather than administrative.

The feasibility of adopting such a system is supported by the Philippines’ ongoing digital transformation. The Department of Information and Communications Technology (DICT) has initiated pilot programs such as e-GOVchain, which integrates blockchain with national identity infrastructure (Mercurio & Hernandez, 2022). Regulatory momentum is evident in the Bangko Sentral’s engagement with virtual asset providers and the development of the Cagayan Crypto Zone, while universities are actively cultivating the necessary talent pool (Brothwell, 2023).

However, technological innovation alone is insufficient. Successful implementation requires more than digitizing existing mechanisms; it demands the integration of local IT infrastructure, capacity building for personnel, and strict adherence to regulatory standards (Valentino & Griarte, 2025; Brothwell, 2023; Azzam et al., 2023). Research stresses that blockchain systems must be designed to improve governance structures fundamentally, rather than merely replicate manual inefficiencies digitally (Azzam et al., 2023).



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1.2 Objectives of the Study

This study aims to develop IskolarBlock, a blockchain-enabled web platform scholarship management system designed to enhance transparency, data integrity, and operational efficiency in Barangay San Miguel's scholarship processes. Specifically, the study seeks:

1. To integrate essential system features that streamline scholarship processing, including secure user authentication, automated document handling through OCR, structured submission of mandatory requirements, administrative dashboards, and real-time application tracking.
2. To implement blockchain technology within IskolarBlock to ensure tamper-proof transaction logging, strengthen data integrity, and provide transparent and verifiable records of scholarship applications and award decisions.
3. To determine the acceptability of IskolarBlock through structured manual testing, focusing on functional performance, usability, stability, and the system's overall readiness for deployment.
4. To evaluate the overall quality and acceptability of IskolarBlock using the ISO/IEC 25010:2023 software quality model.
5. To assess IskolarBlock's contribution to the Sustainable Development Goals, particularly SDG 4 (Quality Education), SDG 9 (Industry, Innovation, and Infrastructure), and SDG 16 (Peace, Justice, and Strong Institutions).

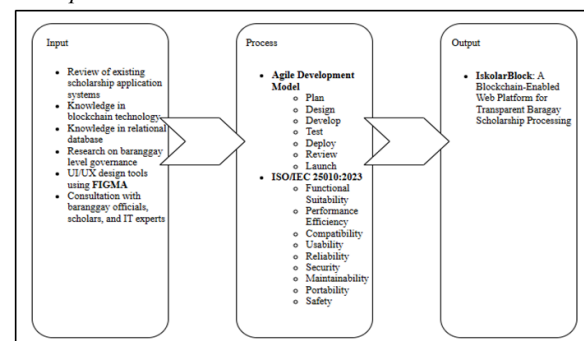
1.3 Significance of the Study

IskolarBlock is a modern, secure, and transparent solution to long-standing challenges in barangay-level scholarship management. By shifting from manual and paper-based procedures to a blockchain-enabled digital platform, the system enhances data integrity, reduces processing delays, and minimizes human errors in document verification. Students benefit from a streamlined application workflow and real-time visibility of their scholarship status, while barangay administrators gain reliable tools for reviewing applications and maintaining tamper-proof records. Through automated OCR, secure authentication, and immutable blockchain logging, IskolarBlock supports accountability, strengthens institutional trust, and promotes fairer educational assistance distribution. Ultimately, the system contributes to more efficient governance and improved access to scholarship opportunities within the community.

1.4 Conceptual Framework or Theoretical Framework

The research is grounded on the theories of Decentralization Theory, Immutable Ledger Theory, Public Value, and E-Governance Theory, and Software Quality Assurance Structure, which altogether form the basis for the design of the IskolarBlock system. Decentralization Theory describes how the distribution of data among various nodes enhances transparency and minimizes the need for central control, while Immutable Ledger Theory guarantees that the scholarship records stored in the ledger cannot be modified. Public Value and E-Governance Theory promotes transparency and efficiency in the delivery of public services, thus enhancing public trust in scholarship management. On the other hand, the Software Quality Assurance Structure, based on ISO/IEC 25010, ensures that the system retains quality, reliability, and usability during the course of its implementation. As shown in Figure 1, the Input-Process-Output (IPO) model served as the foundational conceptual framework for designing and developing IskolarBlock, a blockchain-enabled web platform for transparent barangay scholarship processing. The IPO model was selected for its clarity and effectiveness in illustrating how inputs are transformed into meaningful outputs through structured processes.

Figure 1
Conceptual Framework



METHODS

2.1 Methods and Techniques

A descriptive research design was employed to systematically observe and measure the system's performance and quality, focusing on the perceptions and experiences of different stakeholder groups involved in scholarship management at the barangay level. This method is particularly effective for addressing the what, when, where, and how questions related to the usability, reliability, and transparency of blockchain-based



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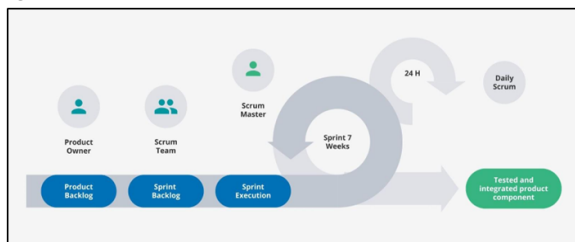
administrative processes (Saunders, Lewis, & Thornhill, 2024).

The study employed cluster sampling, grouping participants based on the natural clusters within the scholarship process. These clusters consisted of the Senior High School student applicants, and College student applicants. This sampling technique ensured that each subgroup was represented proportionally while allowing efficient and organized data collection. The instrument consisted of a structured questionnaire utilizing a 5-point Likert scale to measure key attributes including Functional Suitability, Performance Efficiency, Compatibility, Interaction Capability, Reliability, Security, Maintainability, Flexibility, and Safety.

2.2 System Design and Implementation

IskolarBlock was built using modern web technologies to ensure modularity, maintainability, and responsive UI. The front end used Next.js with TypeScript and Tailwind CSS; form validation relied on Zod and ESLint enforced code quality. Backend services employed Supabase (Postgres, Storage, Auth) for role-based access and file hosting. OCR was implemented with Tesseract and PDF.js; extracted text was piped through n8n for cleaning and organization before being used by the application logic. Blockchain integration used ethers.js to post hashed summaries (keccak256) of applicants' records to a Polygon testnet. JWT tokens secured inter-service communication. Security was considered at architecture level (role-based access, OTP for registration) rather than as an afterthought. System engineering choices and technology mapping are detailed in the project documentation.

Figure 2
Agile Scrum

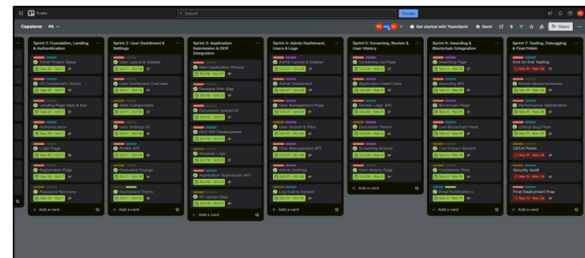


2.3 Development Process

As shown in Figure 2, the Agile Scrum lifecycle guided development across Plan → Design → Develop → Test → Deploy phases. Work was organized into sprints with incremental delivery and end-of-sprint reviews; requirements engineering and continuous stakeholder engagement with Barangay San Miguel shaped feature

priorities and acceptance criteria. Feature Test Case Templates and traceable test artifacts were used throughout testing. As shown in Figure 3, the structure was divided into multiple sprints. This structure breaks down the details of the sprints including deadlines, tasks, categories and progress in an organized and simple manner.

Figure 3
IskolarBlock Trello Project Management Tool



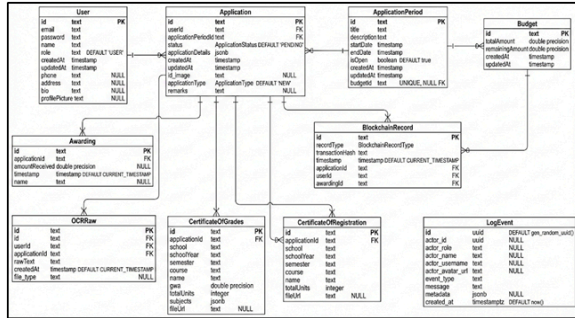
2.4 System Architecture and Design

As shown in Figure 4 the Entity Relationship Diagram, the database schema of IskolarBlock is structured around ten interconnected entities that support the system's core functionality of secure scholarship processing and blockchain integration. The User entity serves as the foundational data point, storing personal identification and authentication credentials (e.g., email, password, role). This entity maintains a one-to-many relationship with the Application entity, allowing a single student to submit applications across different periods. The Application entity acts as the central hub, linking the user to a specific ApplicationPeriod and storing the current status, applicationDetails, and remarks. To manage the scholarship cycle, the ApplicationPeriod entity defines the active timeline (startDate, endDate) and is directly linked to a Budget entity, which tracks the totalAmount and remainingAmount for that specific term. Supporting the verification process, the system utilizes specialized entities for documents: CertificateOfRegistration and CertificateOfGrades store parsed academic data, while the OCRRaw entity archives the raw text extracted from uploaded files for validation purposes. Finally, transparency and accountability are enforced through the BlockchainRecord and LogEvent entities.



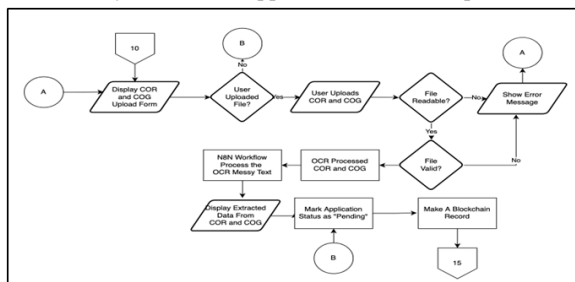
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Figure 4
IskolarBlock Entity-Relationship Diagram (ERD)



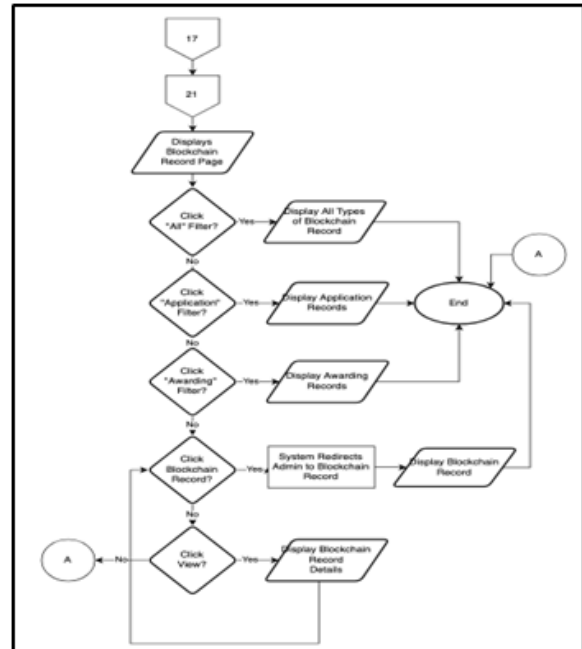
As shown in Figure 5, the Student Applicant Document Upload Flowchart details the technical procedures for submitting academic credentials, specifically the Certificate of Registration (COR) and Certificate of Grades (COG). The process begins at Connector 10, leading to the "Display COR and COG Upload Form" interface. The workflow includes a branching decision node, "User Uploaded File?," to accommodate different submission scenarios. If the user proceeds with an upload (Yes), the system validates the files through two sequential checkpoints: "File Readable?" and "File Valid?". Failure at either stage triggers a "Show Error Message" and redirects the user back to the upload form via Connector A to retry. Upon successful validation, the system initiates an automated extraction sequence: "OCR Processed COR and COG" captures the raw data, which is then refined by the "N8N Workflow" to handle unstructured or "messy" text. The parsed information is presented in the "Display Extracted Data" step for user review. Finally, regardless of whether the path involves new uploads or proceeds via the bypass route (Connector B), the workflow converges to "Mark Application Status as 'Pending'" and generates an immutable "Make A Blockchain Record" before concluding the process at Connector 15.

Figure 5
Flowchart of the Student Applicant Document Upload



As shown in Figure 6, the Flowchart of the Blockchain Data Recording delineates the administrative procedures for accessing, filtering, and verifying the immutable audit logs stored within the system. The process is initiated through Connector 17 (typically from the sidebar) or Connector 21, which triggers the "Displays Blockchain Record Page." From this interface, the system provides robust data retrieval capabilities through a series of decision nodes. Administrators can filter the ledger entries by selecting "Click 'All' Filter," "Click 'Application' Filter," or "Click 'Awarding' Filter." Depending on the selection, the system executes the corresponding retrieval action: "Display All Types of Blockchain Record," "Display Application Records," or "Display Awarding Records" before converging at the End node. Furthermore, the flowchart details specific verification actions. If an administrator chooses to "Click Blockchain Record," the system executes a "System Redirects Admin to Blockchain Record" action (likely linking to an external block explorer). Alternatively, selecting "Click View?" triggers the internal "Display Blockchain Record Details" page, allowing for granular inspection of the hash and timestamp data. Both inspection pathways offer a return loop via Connector A or conclude at the End node, ensuring a flexible auditing workflow.

Figure 6
Flowchart of the Blockchain Data Recording





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2.5 Respondents of the Study

The total respondents were 51, including 1 Barangay administrative staff member, 47 student applicants and 3 IT experts. The Barangay staff member, as a direct implementer of the scholarship process, evaluates the system's integration with existing workflows, administrative usability, and operational flexibility. The 47 student applicants, representing the primary beneficiaries, assess the system's ease of use, transparency, reliability, and overall user experience. Meanwhile the 3 IT experts provide the technical perspective, focusing on functionality, performance and security. The larger number of student participants reflects their central role in utilizing the system and allows for a more comprehensive exploration of usability and satisfaction issues. Both groups used an ISO/IEC 25010:2023-aligned assessment tool to measure the system across key quality attributes. Their combined feedback provided a comprehensive understanding of system quality from both user and expert perspectives.

2.6 Evaluation Strategy

Two complementary evaluation streams were used:

1. Manual testing (acceptability / functional testing): Three iterative test cycles (local and deployed environments) executed a total of 74 test cases covering modules: Authentication & OTP, Profile Management, Document Upload (ID, COR, COG), Scholarship Application, OCR Processing, Blockchain Recording, Admin Dashboard, Reporting (PDF/Excel), Notifications, System Config, UI/UX, Security & Access, and Integration & Performance. Tests were role-based (student vs. admin) and used explicit expected/actual result comparisons recorded in the test template.
2. User quality assessment (ISO/IEC 25010): A questionnaire mapped to the nine ISO/IEC 25010 attributes (Functional Suitability, Performance Efficiency, Compatibility, Interaction Capability, Reliability, Security, Maintainability, Flexibility, Safety) was administered to 51 respondents (47 students, 3 IT experts, 1 barangay staff). Responses were analyzed using descriptive statistics (means, SD) to determine perceived system quality.

RESULTS AND DISCUSSIONS

3.1 Development Outcomes

The implemented system supports: secure OTP registration (duplicate prevention), single-file upload constraints (5 MB limit, png/jpg/pdf), OCR extraction for academic fields (school, semester, GWA, units, school year), admin review interfaces with OCR output, and blockchain logging of two event types (application submission and awarding) storing application ID, user ID, and timestamp hashed via keccak256. Hash creation occurs only after application

completion and upon granting; two separate on-chain records are created when awarding occurs. Access to application statuses is restricted to authenticated users. Minor OCR latency (5–10 s) was observed during testing.

3.2 Test Execution Summary (Manual Testing)

The manual testing phase of IskolarBlock was carried out through three iterative cycles, each demonstrating significant improvements in system stability and performance. During Cycle 1, conducted in the local environment, a total of 76 test executions were performed, yielding 63 passed cases, 4 failed cases, and 9 ongoing cases, resulting in an overall pass rate of 82.89%. Most issues identified in this phase were related to OCR extraction inconsistencies, minor user interface alignment problems, and incomplete module integrations. After addressing these defects, Cycle 2 was executed in the deployed environment, showing marked improvement with a pass rate of 94.74%, indicating that major functional errors including OCR reliability and admin reporting issues had been successfully resolved. The final Cycle 3, conducted in the production environment, achieved near-complete functional stability with a 98.68% pass rate. Only one minor, non-critical user interface issue remained, specifically related to mobile alignment within the admin module, and no test cases were blocked.

Table 1

Cycle 3 Manual Testing Summary

Cycle 3	Number of Test Cases	Percentage (%)
Test Cases: Passed	75	98.68%
Test Cases: Failed	1	1.32%
Test Cases: Ongoing	0	0%
Total Test Cases	76	100%

ISO/IEC 25010 System Quality Evaluation Summary

The system's overall software quality, evaluated using the ISO/IEC 25010:2023 framework, received consistently high ratings across all nine attributes. All criteria scored between 4.76 and 4.91, resulting in an overall mean of 4.85 (Strongly Agree). These results indicate that IskolarBlock performs exceptionally well in functional correctness, efficiency, usability, reliability, and security. High scores in maintainability and flexibility further demonstrate that the system is structurally sound, scalable, and adaptable for future enhancements. The evaluation confirms that IskolarBlock meets international quality standards and is highly acceptable for real-world deployment.



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Table 2
System Quality Evaluation Summary

Characteristics	Grand Mean	Verbal Interpretation
Functional Suitability	4.87	Excellent
Performance	4.80	Excellent
Efficiency		Excellent
Compatibility	4.84	Excellent
Interaction Capability	4.86	Excellent
Reliability	4.86	Excellent
Security	4.85	Excellent
Maintainability	4.87	Excellent
Flexibility	4.91	Excellent
Safety	4.76	Excellent
Final Grand Mean	4.85	Excellent

CONCLUSION AND RECOMMENDATIONS

4.1 Conclusion

IskolarBlock successfully demonstrated that a dual-platform system combining OCR and blockchain techniques can materially improve the transparency, accuracy, and efficiency of barangay scholarship processing. Iterative manual testing and user evaluation using ISO/IEC 25010 criteria confirmed strong system quality, user acceptability, and readiness for operational deployment in the target barangay. The approach reduced manual verification burdens, minimized opportunities for document fraud, and provided auditable, tamper-proof transaction logs, thereby enhancing institutional trust.

4.2 Recommendations

Deployment of the system should proceed through phased adoption in Barangay San Miguel, supported by staff training on blockchain proof verification and OCR review practices to ensure smooth implementation. Future enhancements may include integrating AI-based OCR preprocessing to improve recognition of low-quality images and introducing optional biometric sign-on for stronger user authentication. In terms of scalability and policy considerations, conducting cost-benefit and legal or regulatory analyses is recommended prior to mainnet deployment, along with pilot testing for inter-barangay data

sharing strategies. Further research is also encouraged through longitudinal studies to evaluate impacts on processing time, fraud reduction, and beneficiary satisfaction, as well as exploring Self-Sovereign Identity (SSI) integration to strengthen identity verification mechanisms.

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